

SUNCHEON, Korea, Republic of

WMO: 471740

Lat: 35.0203N

Lon: 127.3694E

Elev: 548

StdP: 14.41

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 19.2	(d) 0.0	(e) 5.6	(f) 28.9	(g) 3.7	(h) 6.8	(i) 29.4	(j) 15.7	(k) 33.0	(l) 14.3	(m) 31.5	(n) 3.7	(o) 320	(p) 0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.9	(c) 90.5	(d) 76.9	(e) 88.3	(f) 76.3	(g) 86.0	(h) 74.9	(i) 79.0	(j) 86.5	(k) 78.0	(l) 84.8	(m) 77.1	(n) 83.2	(o) 5.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	144.2	82.2	76.2	139.7	81.5	75.4	135.7	80.8	43.0	87.0	42.0	84.9	41.1	83.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.8	(b) 12.0	(c) 10.9	(e) 12.5	(f) 94.1	(g) 2.8	(h) 2.2	(i) 10.5	(j) 95.7	(k) 8.9	(l) 97.0	(m) 7.3	(n) 98.3	(o) 5.2	(p) 99.9		
(a) 11.2	(b) 80.2	(c) 2.4	(e) 1.6	(f) 9.4	(g) 81.4	(h) 8.0	(i) 82.3	(j) 6.6	(k) 83.2	(l) 4.8	(m) 84.4	(n) 4.8	(o) 84.4	(p) 84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	32.0	36.5	44.4	53.7	63.5	70.9	77.6	78.4	70.1	58.9	47.4	35.6
	DBStd	16.83	5.97	6.76	6.04	5.64	4.57	3.35	3.83	4.10	4.72	5.41	7.19	6.16
	HDD50	1749	558	382	196	29	0	0	0	0	0	4	132	448
	HDD65	4534	1023	799	640	340	81	1	0	0	9	202	527	912
	CDD50	3887	0	4	21	140	420	627	855	881	602	280	56	1
	CDD65	1195	0	0	0	1	36	178	390	416	161	12	1	0
	CDH74	9310	0	0	0	33	487	1036	3112	3688	910	43	1	0
	CDH80	2695	0	0	0	0	80	186	950	1317	162	0	0	0
Wind	WSAvg	3.9	5.3	4.8	4.8	4.7	4.0	3.3	3.2	3.1	2.5	3.0	3.8	4.8
Precipitation	PrecAvg	56.3	1.2	1.7	2.5	3.6	4.1	6.7	12.2	12.6	6.1	2.2	2.1	1.3
	PrecMax	78.4	3.3	5.0	5.4	9.0	8.5	13.7	25.9	29.6	20.2	7.9	5.8	2.7
	PrecMin	32.7	0.2	0.3	0.6	0.9	1.6	0.8	1.1	4.5	0.9	0.1	0.2	0.2
	PrecStd	11.3	0.6	1.1	1.3	1.9	1.7	3.7	5.4	5.9	4.5	1.7	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.5	61.6	69.6	78.0	85.7	87.6	92.1	94.1	87.8	78.0	71.2	57.3
		MCWB	44.2	56.8	53.0	56.7	63.4	69.3	77.9	77.5	74.6	65.9	61.8	50.4
	2%	DB	48.7	56.9	65.4	73.8	82.1	84.4	89.8	91.6	84.1	74.7	65.8	53.0
		MCWB	41.0	48.6	50.9	55.9	61.5	68.7	77.6	76.8	71.6	62.6	57.6	46.7
	5%	DB	45.1	53.0	61.1	70.2	79.4	82.0	87.7	89.4	81.5	72.4	62.9	49.8
		MCWB	37.8	43.9	49.3	54.6	61.2	68.3	76.9	77.0	70.4	61.5	55.6	44.0
	10%	DB	42.1	48.6	57.1	66.5	76.3	79.8	85.3	87.1	79.2	69.9	59.8	46.3
		MCWB	36.4	41.5	48.7	53.1	60.8	68.1	75.8	76.6	69.1	61.0	53.4	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	58.1	58.5	62.5	69.6	75.4	80.5	80.5	76.8	70.8	64.3	54.0
		MCDB	50.8	60.9	62.2	70.2	72.0	79.8	88.3	88.8	83.8	73.4	67.9	55.5
	2%	WB	42.0	51.9	55.3	60.2	67.5	73.7	79.1	79.2	75.1	68.1	60.9	48.7
		MCDB	46.3	55.3	58.7	66.5	73.2	78.3	86.2	87.0	79.9	72.5	64.1	52.3
	5%	WB	39.9	45.9	52.7	58.3	65.2	72.0	78.1	78.3	73.7	65.3	57.4	44.3
		MCDB	43.9	50.4	59.1	64.1	72.7	77.4	84.6	85.5	78.5	68.2	61.4	48.4
	10%	WB	36.8	42.1	49.3	56.2	63.4	70.5	77.2	77.4	71.9	61.9	54.6	41.5
		MCDB	41.6	47.7	56.4	62.8	71.7	76.7	83.1	83.9	76.3	67.3	58.6	45.8
Mean Daily Temperature Range	5% DB	MDBR	16.3	18.3	21.3	21.8	22.7	16.3	12.2	13.9	16.8	20.8	19.6	16.6
		MCDBR	21.0	25.0	28.6	29.6	30.0	21.6	16.6	18.0	18.9	23.3	23.0	22.1
		MCWBR	13.8	15.7	15.7	13.7	12.1	8.4	5.9	5.9	7.6	11.0	13.6	15.2
	5% WB	MCDBR	19.0	19.5	21.6	19.2	19.9	14.0	14.0	14.7	13.4	13.9	16.9	20.1
		MCWBR	14.3	15.4	15.2	12.5	10.5	7.0	5.6	5.8	6.3	9.1	11.7	15.4
Clear-Sky Solar Irradiance	taub	0.422	0.464	0.538	0.559	0.574	0.629	0.591	0.547	0.486	0.454	0.434	0.419	
	taud	2.002	1.936	1.781	1.785	1.795	1.746	1.897	2.007	2.122	2.082	2.050	2.024	
	Ebn at Noon	234	239	233	237	235	222	229	237	246	241	229	226	
	Edh at Noon	43	52	66	69	69	73	62	55	46	44	41	40	
All-Sky Solar Radiation	RadAvg	825	1058	1387	1588	1731	1519	1395	1442	1289	1171	860	740	
	RadStd	61	88	121	128	185	144	200	192	137	105	89	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	N/A

Nomenclature: See separate page